University of Science, VNU-HCM
Faculty of Information Technology

Introduction to Applied Data Science Course

Designing the Dataset

Le Ngoc Thanh
lnthanh@fit.hcmus.edu.vn
Department of Computer Science

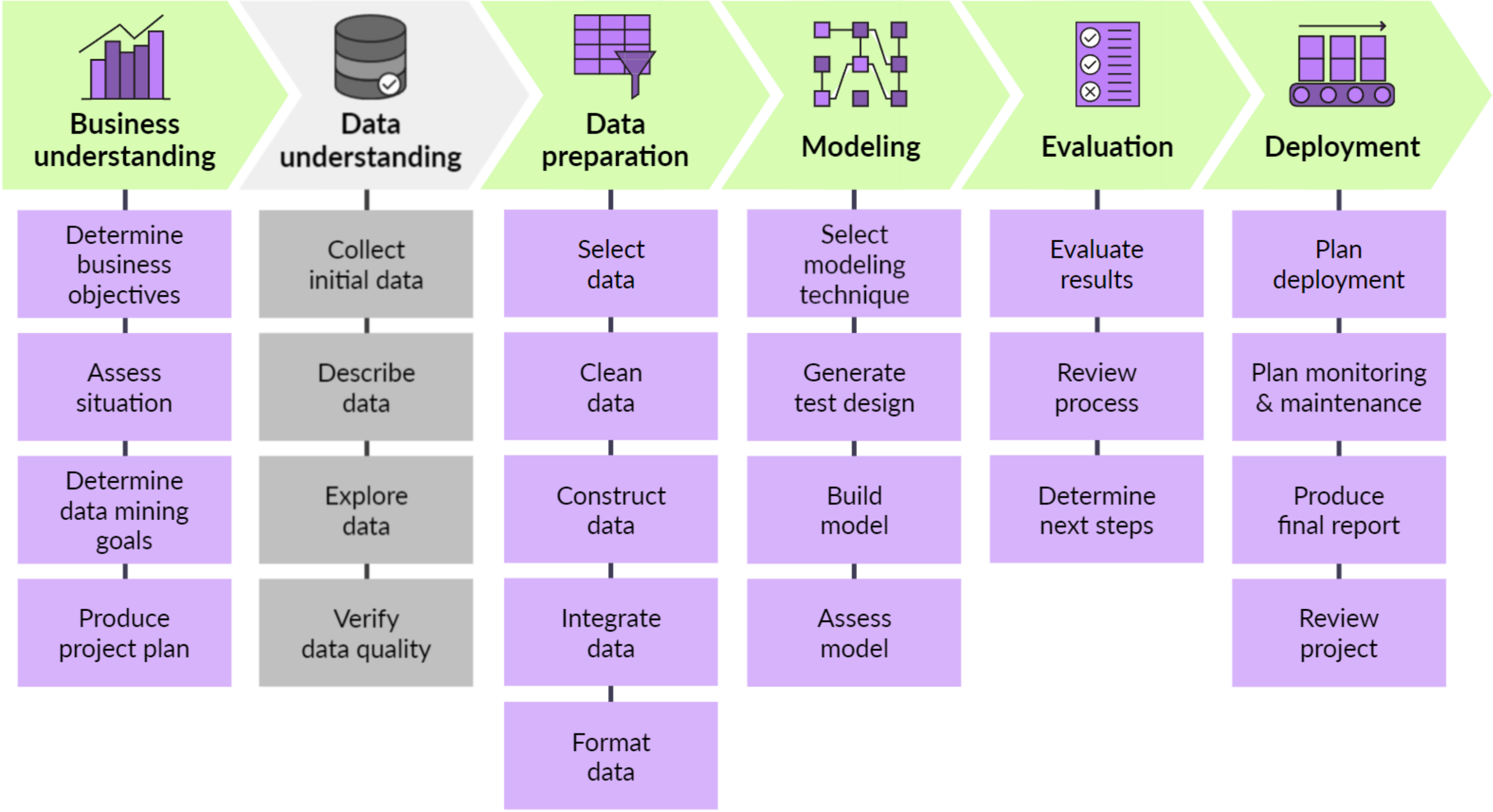
Ho Chi Minh City

Prerequisite of doing data science

- The majority of the time and effort to:
  - Creating
  - Cleaning
  - Updating
- Because:
  - Information from many sources
  - In different formats
  - Data and problems change frequently

Project time

- How long it takes for collecting and preparing data?
  - 79 percent of their time is spent on data preparation (CrowdFlower, 2016)

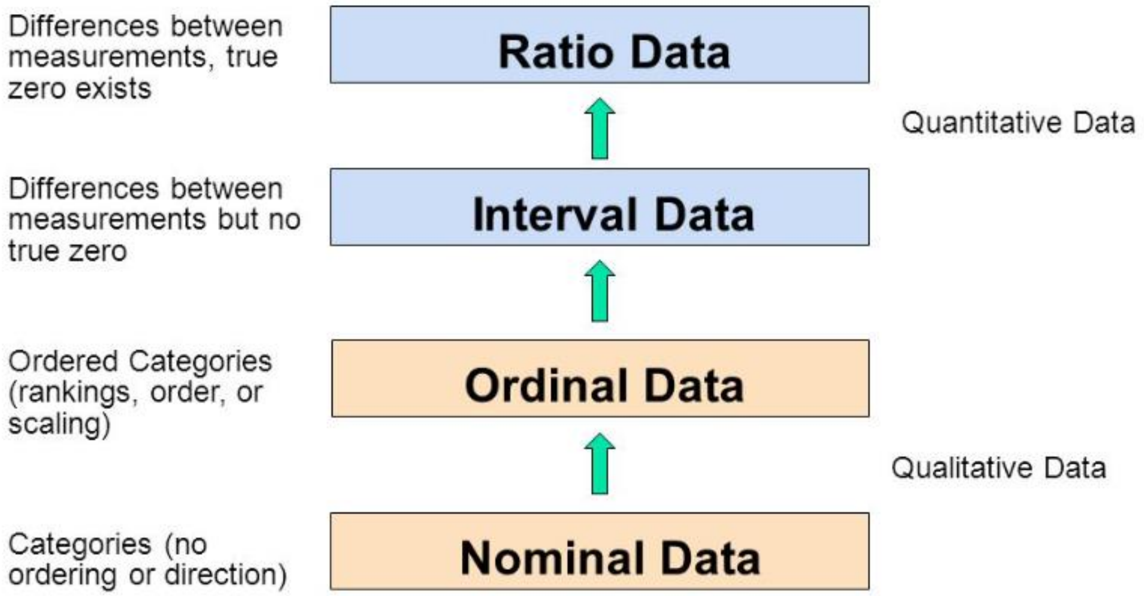


Design the dataset

- Design the dataset:
  - choose attributes from many.
    - Irrelevant or redundant attributes
  - based on our domain knowledge.
    - Iterative process of trial-and-error experiments

Understand and Recognize

- Attribute types:
  - Understanding and recognizing different attribute types is a fundamental skill for a data scientist.
  - The standard types are numeric, nominal, and ordinal.
    - Numeric: interval scale, ratio scale.



Data are always partial and biased

- Data are never an objective description of the world.
  - Data are generated through a process of abstraction, so any data are the result of human decisions and choices.
    - Partial and Biased
  - “A map is not the territory it represents, but, if correct, it has a similar structure to the territory which accounts for its usefulness” (Alfred Korzybski, 1996)
- Therefore,
  - Careful in how we design and gather the data

Structured and Unstructured data

- Structured data are data that:
  - can be stored in a table, and
  - every instance in the table has the same structure (i.e., set of attributes).

  ➔ relatively easy to apply data science to structured data
- Unstructured data are data that:
  - each instance may have its own internal structure
    - For example, collections of human text (emails, tweets, text messages, posts, novels, etc.) as can collections of sound, image, music, video, and multimedia files.

  ➔ difficult to analyze unstructured data in its raw form

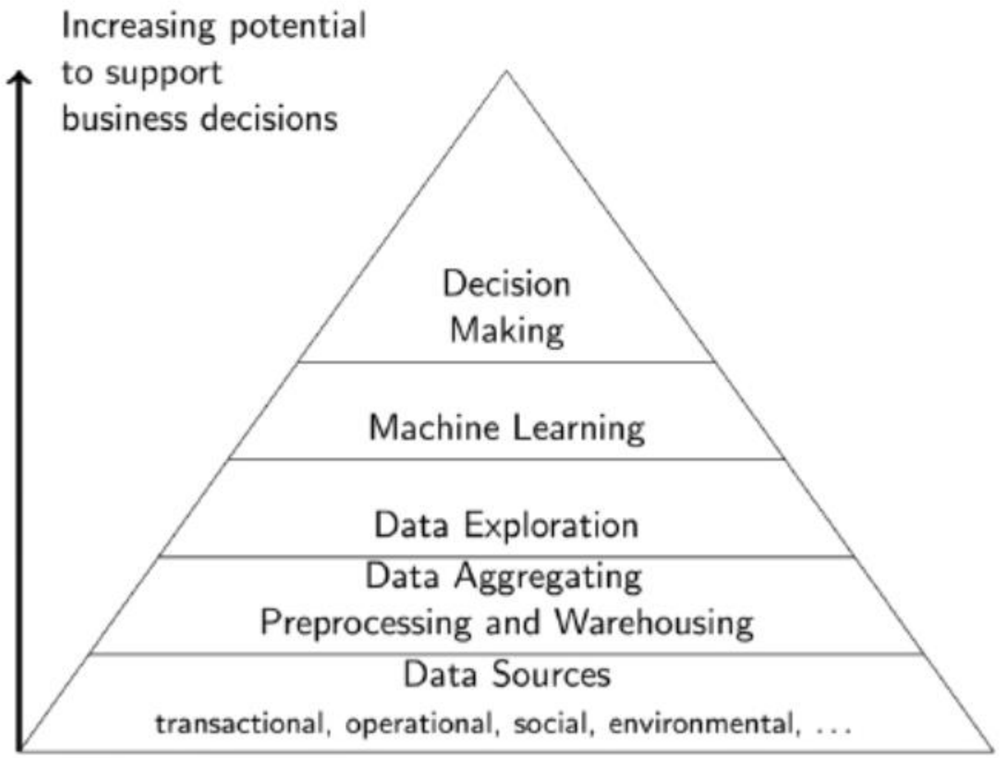
Raw or Derived Attributes

- Attributes can be raw or derived abstractions.
  - Data are derived from an original set of data by applying a function to the original raw data.
  - For example: the average salary in a company, BMI (obesity)

Captured and Exhaust data

- Exhaust data are a by-product of a process whose primary purpose is something other than data capture.
  - For example: social network, amazon website.
  - One of the most common types of exhaust data is metadata.

Data Pyramid and Data Science Pyramid



Initial steps

- Some initial steps after data collection:
  - Data dictionary (understanding data attributes)
  - Data statistics
  - Visualization (begin with each attribute at a time, then in combinations)

The End